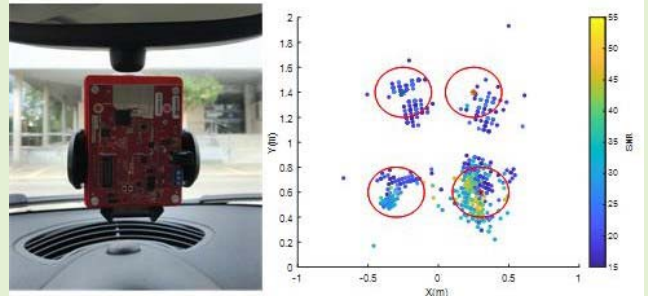


Vehicle Occupancy Detector Based on FMCW mm-Wave Radar at 77 GHz

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Abstract—Millimeter-wave (mm-Wave) radars are a solution in automotive applications, such as advanced driver assistance system (ADASs). This work explores the use of commercially available mm-Wave radar technology for in-cab vehicle applications. A framework for vehicle occupancy detectors based on mm-Wave radar at 77 GHz is presented. A multiple-input-multiple-output (MIMO) frequency-modulated continuous-wave (FMCW) radar from Texas Instruments (AWR1642) is used. The system can detect the presence of people occupying the seats by measuring small movements of the body. To this end, the static clutter is removed; therefore, the point cloud returned by the radar contains information about the body. The radar is installed on the roof of the vehicle to have the maximum field of view of the seats. An algorithm that uses both the density and the dispersion of points around each seat is proposed to determine its degree of occupation. Experimental results have been presented for four- and five-seater vehicles with high accuracy.

Index Terms—Automotive radar, detection, mm-Wave radar, seat occupancy, vehicle-to-everything (V2X).



I. INTRODUCTION

ELECTRONIC systems used to detect, identify, and count occupants within a vehicle are referred to as vehicle occupancy detection systems. Their use is rising since they can help in the mobility field with topics such as decreasing traffic congestion, saving time, and reducing the environmental pollution.

Traffic congestion is rising and becoming a serious issue in most of the cities. It is created by an increase in the number of vehicles on the road, which results in slower speeds, road blockages, longer travel times, wasting valuable time, and negatively impacting the economy.

High-occupancy-vehicle (HOV) lanes are being promoted by government regulations as a new strategy to increase road and city mobility. HOV lanes will serve to alleviate

general congestion by encouraging the usage of carpooling, public transportation, and transit. HOV lanes and congestion toll discount policies are in place to promote vehicle sharing. However, vehicle occupancy detection, needed to implement such policies, is often done using labor-intensive manual procedures [1]. Increasingly, access to congested cities and places is being gradually restricted, taking into account environmental criteria. As a result, the demand for vehicle occupancy detection systems is expected to rise in the coming years.

Radar technology can identify the presence of a person even in the most difficult of environments, such as in bad weather conditions or in the presence of intense light or darkness. Millimeter waves (mm-Waves), unlike other sensing technologies, can pass through materials such as plastic and clothing. In this way, sensors can be hidden behind housing or placed inside or beneath other components of the car, thus being noncontact and nonintrusive. The pressure sensor, for example, cannot distinguish the difference between a human and a static object, and cameras will not detect a baby in bright or dark light conditions. Recently, the authors have proposed a seat-occupancy detector and breathing monitoring based on low-cost coherent-pulsed mm-Wave radar at 60 GHz. However, due to its short range (due to its low transmitting power, the detection range is of the order of 1 m), a radar sensor must be installed in front of each seat [2]. Ultrasonic range sensors have also been proposed for seat occupancy [3]; however, they have

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similar problems as pulsed radar. To avoid reflections from adjacent objects or passengers, one must be installed for each seat. In addition, ultrasonic sensors can suffer interferences from sound sources in the vehicle, or they can be audible, depending on the frequency, by animals such as pets [3]. This study proposes a different solution that overcomes those limits. It proposes a frequency-modulated continuous-wave (FMCW) multiple-input-multiple-output (MIMO) radar operating at the 77-GHz band to determine the degree of occupancy of vehicles with multiple passengers.

This article is organized as follows. After the introduction, in Section II, different systems proposed to detect occupants inside a vehicle are reviewed. Section III describes the proposed solution based on 77-GHz mm-Wave radar. Experimental results and discussion are given in Section IV. Finally, conclusions are provided in Section V.

II. STATE OF THE ART

Several studies and companies have attempted to design an automated detection method to overcome this challenge. Vehicle occupancy detection methods proposed in the literature are mainly based on video systems, which have different ways of classifying images [1], [4], [5], [6], [7], [8], [9], [10] and thermal images [11], [12], [13], [14]. Recently, commercial products based on video and computer tools have been designed for these applications and marketed by suppliers such as Xerox, Conduent Inc., Invision AI, Fortran Traffic Systems Limited, Indra, and Siemens. However, since these products are not boarded in the car but are external, and due to various factors such as tinted windows, obstruction and variation in occupant posture, vehicle speed, size, shape, imaging geometry variation, and bad weather conditions, the vehicle passenger occupancy detection method based on video has several drawbacks in distinguishing between passengers and objects placed on the seat, which can reduce cases of successful detection. Systems based on thermal images have a higher cost compared to optical video-based systems. Thus, the design of alternative cost-effective vehicle occupancy detection systems is needed to attract more customers.

The integration of detection devices inside vehicles is another option to determine their degree of occupancy. On the other hand, the use of onboard seat-occupancy detection systems is also used in other applications, so one only system could be shared by these applications. For instance, airbag systems are fundamental safety elements in modern vehicles, and they require information about the presence and type of passenger who may occupy the seat to work properly. Unwanted airbag activation results in costly repair operations. In addition, activating an airbag in a seat occupied by a rear-facing baby seat can result in fatal injuries [15]. As a result, sensors to detect seat occupancy are required, preventing the airbag from activating when the seat is not occupied by a user or by a rear-facing baby. Seat belt is another mandatory safety system. A different application of the seat occupancy detector is as a seat belt reminder. Conventional seat belt reminder systems work with weight sensors. As a result, when the seats are occupied by luggage or purchases, they generate unwanted alarms. Seat occupancy monitoring systems can also

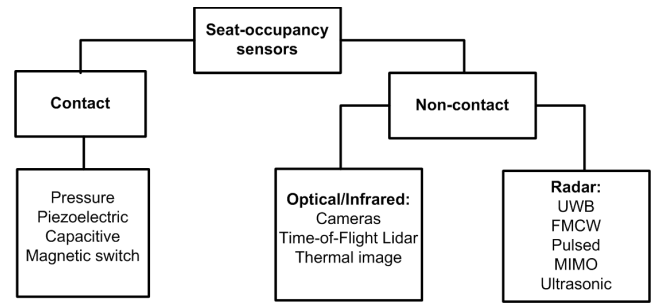


Fig. 1. Classification of seat-occupancy detection systems.

be used to improve passenger's comfort, in addition to meeting regulatory requirements, such as managing the air conditioning system based on the occupancy rate. Another use is to alert the presence of children or pets, preventing the driver from forgetting them inside the vehicle.

In the literature, various approaches to onboard detect seat occupancy have been proposed. There are two types of methods: with contact and without contact. The first approach is based on installing a sensor in the seat to detect the presence of a person, while the second way relies on devices that detect the presence of an occupant at a distance using electromagnetic waves or optical systems. The diagram shown in Fig. 1 summarizes different seat-occupancy detection systems.

Pressure sensors are a widespread contact method to detect seat occupancy [16], [17]. The main limitations of these sensors are their sensitivity to vibrations, and the passenger must be in contact with the seat. Another type of sensor is based on the use of capacitive sensing and consists of a set of electrode arrays embedded in the seat [18], [19], [20]. On the other hand, these sensors are sensitive to interference and the distance between the body and the seat.

Recently, contactless seat-occupancy detection is gaining interest. Optical sensors, such as the use of cameras [21], [22], [23], could be a viable alternative as detection systems, particularly for rear-seat applications. Recently, in [23], a camera-based method to monitor breathing from a reflective object attached to the belt has been studied. The time-of-flight measured with light detection and ranging (LiDAR) sensors has also been proposed as a seat-occupancy detection method [24] based on a Kalman filter. The main disadvantage of camera-based systems is that they rely on face or shape detection, making it difficult to use in low-light situations or when children are dressed. Unlike camera-based vision systems, LiDAR provides distance information. However, the price of LiDAR sensors is still high (compared with integrated mm-Wave radars), and the computational cost for image processing is high. Infrared cameras can detect people in low-light conditions thanks to body heat, but humans may not be distinguished when the interior of the automobile is hot, such as in summer. In addition, infrared cameras are more expensive than traditional ones. These vision systems use extensive signal processing techniques, which greatly increases the computational cost and resources that must be added to that of the system. Recent developments in new technologies, such as mm-Wave radar sensors and vehicle-to-everything (V2X) communications [25], can contribute to developing future vehi-

cle occupancy detection systems. The cost of these radars is falling due to increasing mass marketing in different advanced driver assistance systems (ADASs) and the development of mm-Wave semiconductor devices for 5G communications systems. Radar systems applied to vehicle occupancy detection are beginning to be investigated. Car designers have already successfully integrated mm-Wave sensors within the vehicle cabin for automotive applications. One of these potential applications is the ability to detect occupants within the vehicle considering a variety of both lighting conditions and sensor locations, regardless of movement. For instance, this can help automotive systems detect an unattended child in the rear of the car or the position of the occupants to control the temperature.

Recently, different radar types have been studied for vital-signs monitoring, such as [26], [27], and [28], continuous-wave (CW) Doppler radars [29], [30], [31], FMCW radar [32], or impulse radio ultrawideband (IR-UWB). However, a relatively small number of works have focused on automotive in-cabin applications. Seat occupancy based on FMCW microwave and mm-Wave radars at 24 and 77 GHz has been recently proposed in [33] and [34], respectively. Seat occupancy detection based on monitoring the received UWB signal has been proposed in [35]. A system based on a coherent pulse radar at 60 GHz capable of detecting the presence of a passenger and measuring the breathing rate has been presented in [2]. This work is based on a time-division multiplexing (TDM) MIMO FMCW radar technology because it is able to detect small movements, as well as the range and angle of multiple targets simultaneously [36]. These characteristics, in addition to the maturity of this technology in the automotive sector and, therefore, the experience of vehicle manufacturers, make it ideal for this application compared to other radar technologies.

III. SYSTEM DESIGN AND THEORETICAL BACKGROUND

A. System Overview

The system is based on the mm-Wave AWR1642 radar chipset [37] from Texas Instruments (TI, Dallas, TX, USA). It incorporates an FMCW radar that has two transmitters and four receivers with a built-in phase-locked loop (PLL) and analog-to-digital converters (ADCs). TI's evaluation board, AWR162 BOOST [38], which also integrates a C674x digital signal processor (DSP) and an ARM R4F baseband microcontroller for the signal processing, the radar setup, and communications with an external host controller, is used in this work. Fig. 2 shows a block diagram of the board. The radar can operate in the 76–77- or 77–81-GHz band transmitting up to 12.5 dBm. The board implements an interleaved virtual antenna array [36] with eight equivalent antennas spaced $\lambda/2$. Each of the antennas of the array has a maximum gain of 9 dBi in the 76–81-GHz operating band.

The radar is suspended from the sunroof of the car for demonstration purposes, pointing toward the seats, as seen in Fig. 3. In a real nondemonstration scenario, it will most likely be installed around the rearview mirror, dashboard, or even

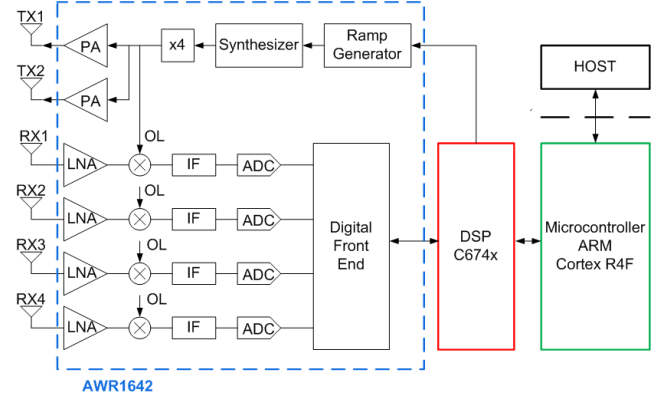


Fig. 2. TI evaluation board AWR1642 BOOST diagram.

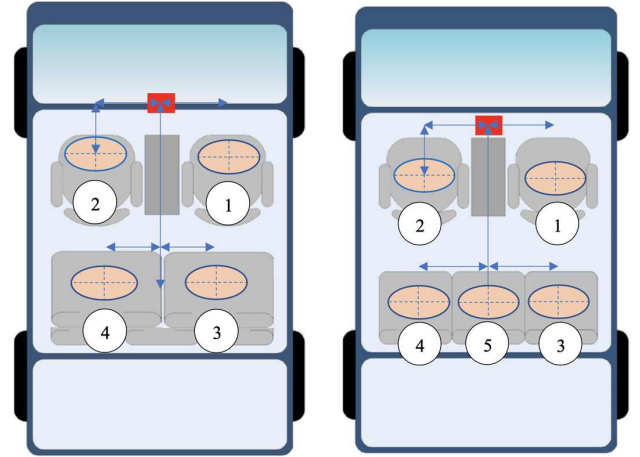


Fig. 3. Installing the radar in a four-seat car, a five-seat car, and seat numbering.

on the ceiling. In the mounted position, the radar must have maximum visibility in the direct line of sight of the seats to avoid blocking the signal. In addition, the system can be applied in gesture recognition applications as proposed in recent works [39]. Two cases will be investigated, a four-seater car and a five-seater car, as shown in Fig. 3. Fig. 4 shows the radar board installed on the front glass of a vehicle that was used for testing in this work.

B. Basic Measurement Theory

An FMCW radar transmits a frame composed of L chirp signals (see Fig. 5). Each chirp is a signal whose frequency varies with time from the minimum value $f_{\min}$ to a maximum value $f_{\max}$ following the shape of a saw tooth, being $B = f_{\max} - f_{\min}$. A linear chirp signal performs a linear sweep in frequency from a specific frequency value to a higher one (or vice versa). A typical FMCW chirp signal can be written as

$$x_c(t) = \exp\left(j2\pi\left(f_{\min}t + \frac{\mu}{2}t^2\right)\right), \quad 0 \leq t \leq T \quad (1)$$

where $\mu = B/T$ is the slope of the chirp signal, with T being the sweep time. The transmitted signal is composed of



Fig. 4. Photograph of the radar attached to the front glass for testing.

L chirps, and it can be expressed as

$$x_T(t) = \sum_{l=0}^{L-1} x_c(t) \prod \left(\frac{t-lT}{T} \right) \quad (2)$$

where $\prod(t)$ is the normalized rectangular signal (between $t = 0$ and $t = T$). In each of the four receivers, the received signal and transmitted signals are mixed and filtered to, finally, obtain the beat signal, which is sampled and discretized at a rate of f_s using a fast ADC for the postprocessing. The received signal of the k th array element for each l th chirp is given by [40]

$$x_{r,lk}(t) = ax_c(t - \tau) \exp(j2\pi f_D l T) \exp \left(j \frac{2\pi}{\lambda} dk \sin \theta \right). \quad (3)$$

The beat signal obtained for the l th chirp and the k th array element $y_{lk}(t)$ can be expressed as [40]

$$y_{lk}(t) = x_{r,lk}(t) \cdot x_T(t)^* \quad (4)$$

$$y_{l,k}(t) = a \cdot \exp \left(-j \left(2\pi f_{\min} \tau - \mu \tau^2 / 2 \right) \right) \cdot \exp(-j2\pi \mu \tau t) \cdot \exp(j2\pi f_D l T) \cdot \exp \left(j \frac{2\pi}{\lambda} dk \sin \theta \right) \quad (5)$$

where a is the amplitude of the target, d is the spacing between adjacent antenna elements, λ is the wavelength at the carrier frequency, τ is the propagation delay, f_D is the Doppler frequency shift due to velocity, and θ is the angle of arrival (AoA). Therefore, the first term is a complex amplitude; the second is a time-of-arrival term that depends on the range; the third is the Doppler and it is related to movement; and the last term is the direction of arrival (DOA) that depends on the AoA. The samples $y_{lk}[n] = y_{lk}(nT_s)$ for $n = 0, 1, \dots, N_s - 1$ are saved in a tridimensional matrix.

In the case of a stationary target, the propagation delay τ is proportional to the distance to the target r

$$\tau = \frac{2r}{c}. \quad (6)$$

However, in the case of a person, there is a periodic modulation caused by the displacement of the chest due to breathing that can be modeled as follows:

$$r = r_0 + r_b \cdot \sin(2 \cdot \pi \cdot f_b \cdot t) \quad (7)$$

where f_b is the breathing frequency (typically between 0.1 and 0.3 Hz) and r_b is the breathing displacement of order of 5 mm [41]. Replacing τ in (5), the following expression is obtained under some reasonable approximations:

$$y_{l,k}(t) \approx a \cdot \exp(-j(2\pi f_{\min} 2 r_b / c \cdot \sin(2 \cdot \pi \cdot f_b \cdot t))) \cdot \exp \left(-j \left(2\pi f_{\min} \tau_0 - \mu \tau_0^2 / 2 \right) \right) \cdot \exp(-j2\pi \mu \tau_0 t) \cdot \exp(j2\pi f_D l T) \cdot \exp \left(j \frac{2\pi}{\lambda} dk \sin \theta \right) \quad (8)$$

where $\tau_0 = 2r_0/c$ is the mean propagation delay associated with the distance to the body, r_0 . Since the chest displacement is below the range resolution (of the order of few centimeters), the last equation has been approximated by the following expression:

$$\exp(-j2\pi \mu \tau t) \approx \exp(-j2\pi \mu \tau_0 t). \quad (9)$$

Furthermore, quadratic terms, such as $\mu(r_b/c \cdot \sin(2 \cdot \pi \cdot f_b \cdot t))^2$ and other mixing products involving the breathing amplitude, can also be neglected as the first approximation.

Analyzing the first term of (8), it can be seen that the phase of the received signal is modulated by the slow variations introduced by breathing

$$\phi(t) = -2\pi f_{\min} 2 r_b / c \cdot \sin(2 \cdot \pi \cdot f_b \cdot t) = -\frac{2\pi}{\lambda_{\min}} 2 r_b \cdot \sin(2 \cdot \pi \cdot f_b \cdot t). \quad (10)$$

Chest displacements are comparable to the wavelength at the lowest frequency $\lambda_{\min} = 3.89$ mm. Therefore, important phase changes in the slow time will be expected. These phase changes do not affect the determination of position or velocity because they are below the range and velocity (Doppler) resolutions, and the phase variation due to breathing is approximately constant in each chirp. However, if these microvariations as a function of time between consecutive radar acquisitions are considered, then information about the target's breathing signature can be retrieved [42], [43]. Particularly, in this work, it is used to discern between persons and other objects.

The processing chain implemented in the board DSP [44] is described in Fig. 6. A windowed fast fourier transform (FFT) applied to the samples of each chirp (FFT in range) is used to estimate the range. The resolution in range depends on the ability to discriminate the peaks of the spectrum to resolve targets. Thus, the range resolution is determined by

$$\Delta R = \frac{c}{2B}. \quad (11)$$

In order to achieve the maximum range resolution, the maximum sweep bandwidth was chosen in the radar settings. Considering a bandwidth of 3.4 GHz, a range resolution of 4.4 cm from (11) is obtained. The main goal of this work is

TABLE I
CONFIGURATION OF THE FMCW RADAR

Parameter	Value
Start Frequency	77 GHz
Sweep bandwidth	3.44 GHz
Sweep slope	70 MHz/ μ s
Sweep time	966 μ s
Frame periodicity	100 ms
Sampling frequency	5209 kbps
Number of samples per chirp	256
Number of chirps per frame	32
Transmit antennas	2
Receive antennas	4
CFAR method	Cell Averaging
Noise averaging window length in range direction	8
Guard length in range direction	4
Noise averaging window length in Doppler direction	4
Guard length in Doppler direction	2
Detection threshold factor	15 dB

to distinguish between people and inanimate objects that may be in the seats or between people and static clutter based on the movements detected that could be as small as those associated with breathing. Thus, before performing the 2D-FFT, a static clutter removal algorithm is implemented by subtracting the mean value of the input samples. For each range bin, per each antenna, the mean value of the samples is calculated and subtracted from the samples. The frame periodicity has been chosen to have a scene update rate of 10 fps allowing slow variations to be tracked. With the parameters listed in Table I, the maximum unambiguous range is 11.1 m, the maximum radial velocity is 1 m/s, and the radial velocity resolution is 0.0625 m/s.

Subsequently, another FFT (Doppler FFT) is applied for each chirp and antenna channel, obtaining a range-Doppler map for each k th array element. In order to reduce clutter, a constant false alarm rate (CFAR) algorithm is used to estimate a threshold. Points below this threshold are considered clutter and are removed from the range-Doppler map. Several CFAR algorithms have been proposed in the literature so far [45], [46]; however, in this work, the cell-averaging (CA-CFAR) is used for its simplicity [47]. This algorithm estimates the surrounding noise power by averaging the neighboring cells to the cell of interest. The parameters of the CFAR algorithm used in the experiments have been included in Table I. In this work, eight training cells and four guard cells on each side of the cell under test (CUT) in range, and four training cells and two guard cells on each side in the Doppler direction were considered. The threshold is obtained from the averaged power noise and a threshold factor or minimum signal-to-noise ratio (SNR) of 15 dB. The third FFT (AoA FFT) is then applied [44], [48]. The FFT-based algorithm for DOA detection is the most widely used algorithm because of its low complexity and ease of implementation [49]. The angle estimation is performed by processing the received signal at the array composed of multiple elements. The simplest algorithm is the angle FFT algorithm. If a Fourier transform is performed in the spatial dimension through the receiver elements (known as angle FFT), it will be possible to distinguish objects based on their AoA (θ) in azimuth.

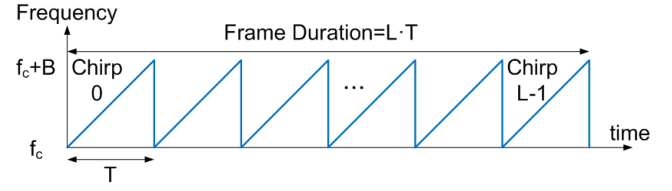


Fig. 5. Frame of chirps.

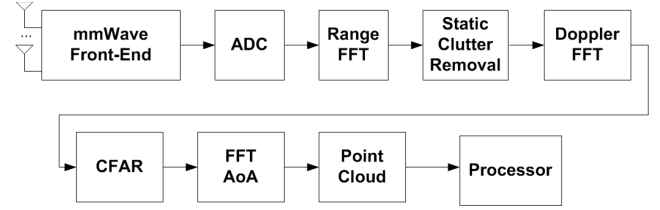


Fig. 6. mm-Wave radar processing chain.

In this application, angular accuracy is not critical because it is used to estimate approximately the region in which the passengers are located. Therefore, the angle FFT algorithm is used in this work. However, enhanced angular algorithms can be employed, such as ESPRIT [50], MUSIC [49], [51], or Capon beamformer [52]. In this work, the point cloud data provided by the radar are used as input to the occupancy detection algorithm. The total procedure requires three FFTs (in range, velocity, and angle), as shown in Fig. 6. The mm-Wave SDK 3.5 firmware [44], loaded in the internal DSP unit in the radar board, was used to calculate the point cloud. The radar is configured using command-line interface (CLI) commands, and the measured data are returned by the USB bus. The data are updated at a rate of 10 fps, which is enough to detect variations in the point cloud caused by breathing. On the radar sensor board, preliminary processing of the signal received by the radar is performed, yielding a point cloud [44], [48]. The board provides an SNR for each point estimated from the previously saved noise profile. The point cloud is then processed on a Raspberry Pi 4 or a personal computer. The seat occupancy algorithm is described in Section IV.

IV. SEAT OCCUPANCY ALGORITHM

Figs. 7 and 8 show the point cloud and the measured SNR for each point during 10 s in a car with four and five seats, respectively. The radar only returns the points whose SNR value is higher than the threshold (15 dB in this case). Therefore, points below this threshold do not appear on these plots. The regions corresponding to empty seats generally do not show the presence of dots. In this situation, these points are treated as static clutter, which is eliminated with the algorithm specifically designed for this function. In order to see the effect of this algorithm, Fig. 9 shows measurements for an empty four-seater car applying or not applying the static clutter algorithm. The threshold factor determines the desired probability of a false alarm. The value was chosen empirically from measurements in a vehicle with no occupants. It was found that, for most situations and vehicles, a level of 15 dB eliminates reflections from stationary objects in the region of

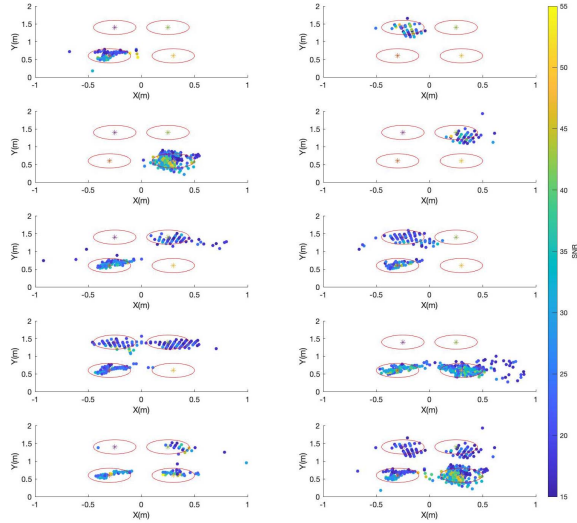


Fig. 7. Point cloud measured for a four-seater car depending on the seat. The colored bar shows the measured SNR. The seating areas are marked in red.

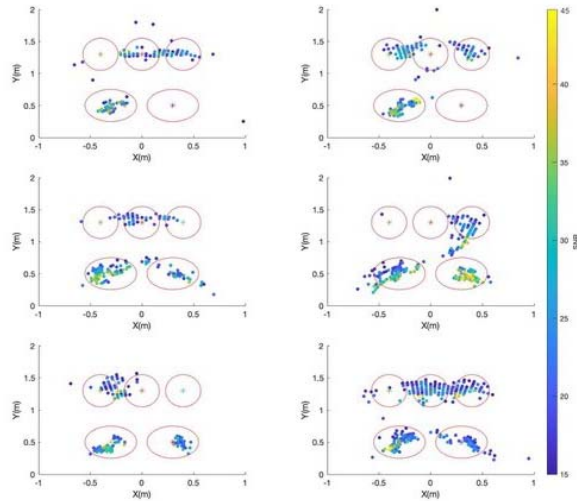


Fig. 8. Point cloud measured for a five-seater car depending on the seat. The colored bar shows the measured SNR. The seating areas are marked in red.

interest [see Fig. 9(b)]. Points whose range is greater than the seat region are removed because they are assumed to be caused by reflection from vehicle walls or from an object outside the vehicle. Based on these plots, an algorithm for classification from some features has been developed.

First, some parameters have been defined to roughly determine the seating areas shown in red in Figs. 7 and 8. The area of space occupied by a body approximates that of an ellipsoid. These ellipsoids are referenced to the coordinates of the center of the interior of the vehicle (X_k, Y_k) and the radius in horizontal (R_{xk}) and vertical (R_{yk}), where k is the index of the seat, $k = 1, \dots, N_{\text{seats}}$. These parameters are the input of the algorithm and can vary depending on the car. However, they can be obtained from a table as a function of the vehicle model or specified manually by the user. As an example,

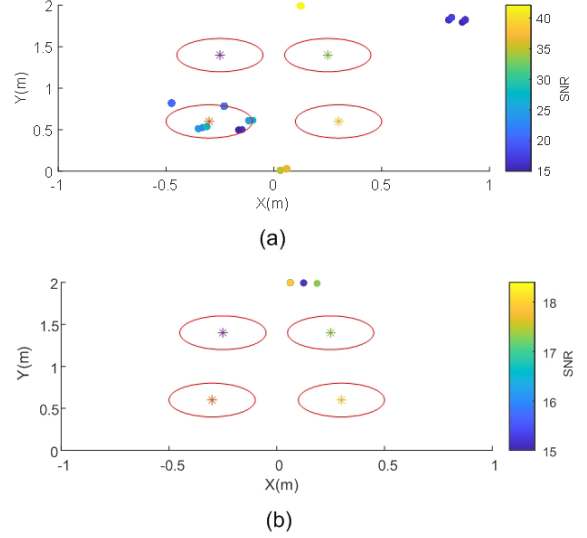


Fig. 9. Point cloud measured for a four-seater car without passengers when the static clutter removal algorithm is (a) not applied and (b) applied. The colored bar shows the measured SNR. The seating areas are marked in red.

TABLE II
GEOMETRIC PARAMETERS ACCORDING TO THE NUMBER OF SEATS IN THE VEHICLE

Parameter	4 seats	5 seats
Distance radar to front seat row	60 cm	50 cm
Distance between front seats	60 cm	60 cm
Distance between back seats	60 cm	80 cm
Distance between front and back seats	80 cm	80 cm
Horizontal semi-axis front seats	20 cm	25 cm
Horizontal semi-axis back seats	20 cm	17 cm
Vertical semi-axis front seats	20 cm	20 cm
Vertical semi-axis back seats	20 cm	20 cm
Threshold (TH) for front seats	0.01	0.01
Threshold (TH) for back seats	0.005	0.005

Table II provides the values that define the seating areas for two cars with four and five seats.

The algorithm takes into account the points cloud saved over a period of time (T_a). Once the regions of every point have been calculated, their coordinates are saved in the vectors $\bar{X}_k$ and $\bar{Y}_k$. A point (x_i, y_i) is in zone k if it belongs to the ellipsoid

$$\sqrt{\left(\frac{x_i - X_{Ck}}{R_{xk}}\right)^2 + \left(\frac{y_i - Y_{Ck}}{R_{yk}}\right)^2} < 1 \quad (12)$$

where X_{Ck} and Y_{Ck} are the coordinates of the center of the area k , and R_{xk} and R_{yk} are the semiaxes of the ellipsoid in the directions x and y , respectively.

Then, the number of the point N_k of the cloud that belongs to a specific area is the length of vector $\bar{X}_k$ or $\bar{Y}_k$.

The mobility of a seated person produces a dispersion of the points in each occupied seat. The standard deviation of the points in each zone gives a measure of this dispersion

$$\sigma_k = \sqrt{\frac{1}{N_k} \sum_{i=1}^{i=N_k} (X_{ki} - E(\bar{X}_k))^2 + (Y_{ki} - E(\bar{Y}_k))^2} \quad (13)$$

where $E()$ denotes the mean operator, and X_{ki} and Y_{ki} are the components of the vectors $\bar{X}_k$ or $\bar{Y}_k$, respectively. The following classification function is defined:

$$f_k = \frac{\sigma_k \cdot \frac{N_k}{N}}{\sum_{k=1}^{N_z} \sigma_k \cdot \frac{N_k}{N}}. \quad (14)$$

This normalized function corresponds to the standard deviation of the points in each zone (σ_k) multiplied by the density of points in the zone (N_k/N). The function includes the fraction of points to avoid false detection when a seat is empty, characterized by a small number of points read by the radar, sometimes dispersed due to residual clutter. A threshold is established for each zone, TH_k . If $f_k > TH_k$, the seat is considered occupied. The seat state (0 or 1) is saved in a vector so_k

$$so_k[n] = \begin{cases} 1, & f_k[n] > TH_k \\ 0, & \text{else.} \end{cases} \quad (15)$$

A low value of threshold TH_k increases the probability of false alarms. It has been obtained experimentally. The threshold value varies between the front and rear seats due to certain blockage; therefore, the number of points in the rear regions has been observed smaller than for the front zones. The threshold values are fixed and are obtained from the typical values obtained for the function f_k from sets of training measurements. For this purpose, the mean values and standard deviations of the f_k function for empty seats are calculated considering various scenarios that include different numbers of seats occupied by different volunteers. The threshold is then determined as the mean value minus three times the standard deviation. The values obtained are listed in Table II. It has been found that, in most cases, the same values can be used even if the vehicle is different. The values obtained from the training measurements have been statistically validated with independent measurements, and the results of the confusion matrix will be presented in Section V.

Finally, the seat occupancy vector is averaged to eliminate errors associated with passenger movement (e.g., when entering or exiting the vehicle). A moving average filter is used (e.g., by averaging the last $M = 3$ samples), and the result is compared to a threshold ($m = 0.5$). The filtered seat occupancy $\overline{so}_k$ is obtained from

$$\overline{so}_k[n] = \begin{cases} 1, & \frac{1}{M} \sum_{i=n-M+1}^n so_k[i] > m \\ 0, & \text{else} \end{cases} \quad (16)$$

where so_k is the seat occupancy for seat k th.

Fig. 10 shows the flowchart of the algorithm. The algorithm has been implemented in both MATLAB and Python. The latter allows running the algorithm on different platforms including Raspberry Pi platforms. Since the proposed algorithm does not require managing vectors with long length or complicated functions, the computational cost and memory capacity is low. The evaluation of the classification function only needs few milliseconds, that is, noticeably lower than the acquisition time required to collect a reliable point cloud with enough number of points. Therefore, the rate at which the algorithm

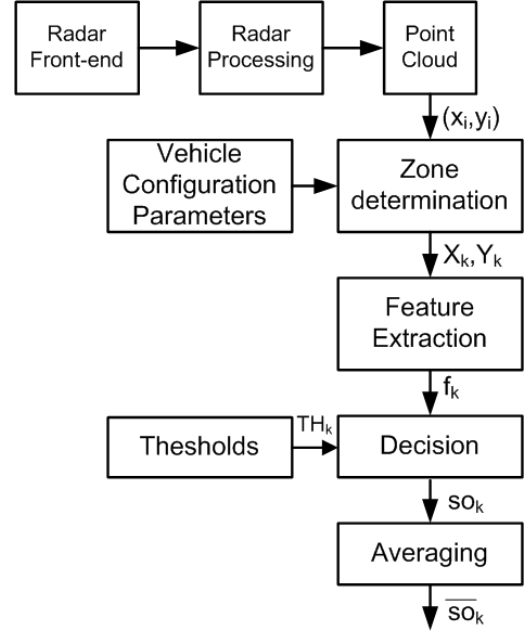


Fig. 10. Processing chain flow diagram.

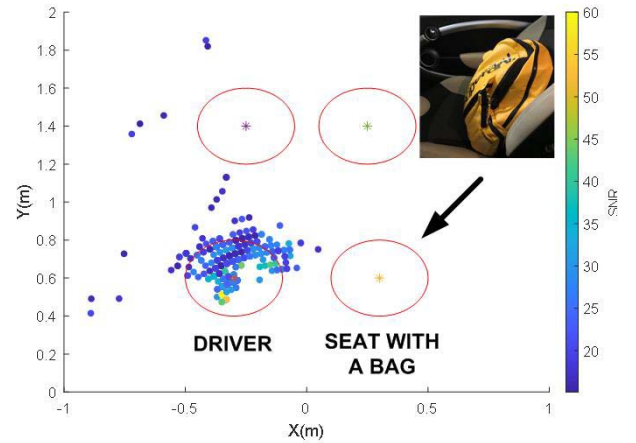


Fig. 11. Point cloud measured with the driver present and a bag (inset figure) occupying the passenger seat.

can return seat occupancy depends on the acquisition time parameter. For commercial products, the algorithm can be easily implemented on other embedded devices even in the microcontroller embedded in the radar itself.

V. RESULTS

A. Occupancy Detection in Different Situations and Influence of Algorithm Parameters

First, the acquisition radar algorithms have been tested to distinguish between people and objects that may occupy the seats. Fig. 11 shows the point cloud measured in a car occupied by the driver and a bag in one of the side seats. The bag contains a laptop to present a high reflectivity in order to increase the level of the echoes collected by the radar thus considering the worst case. It can be seen that there are practically no points on the seat on which the bag is located. The reason is that the bag remains stationary and the signal is

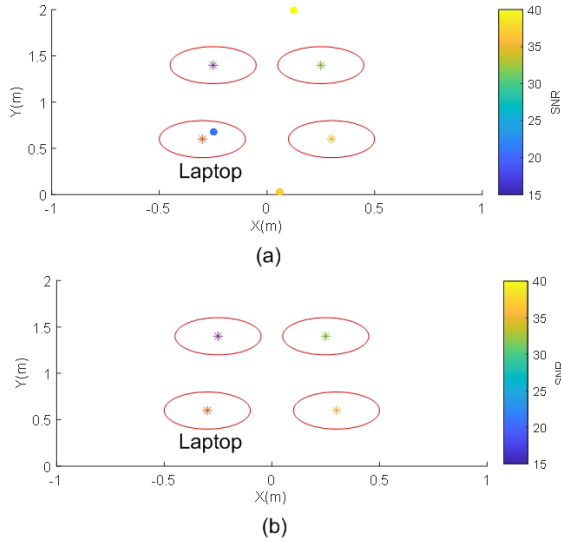


Fig. 12. Point cloud measured for a four-seater car without passengers with a laptop in one seat (a) without the static clutter removal algorithm and (b) with this algorithm activated. The colored bar shows the measured SNR. The seating areas are marked in red.

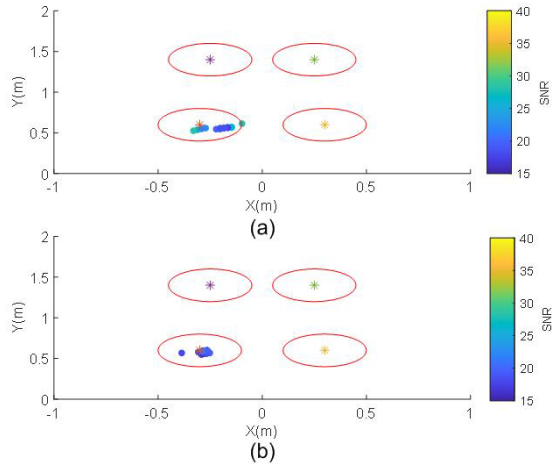


Fig. 13. Point cloud measured with a stationary driver: (a) breathing and (b) without breathing.

canceled by the static clutter suppression algorithm. In order to confirm this fact, a measurement is performed inside the car in which there is only the static clutter algorithm. The results are shown in Fig. 12. A stationary strong point is measured due to the presence of the metal case and other metal objects but is eliminated by the clutter removal algorithm. This figure confirms the correct selection of the detection threshold in the CFAR algorithm.

Breathing produces a low-frequency micro-Doppler due to the periodic movement of the chest. Consequently, it is possible to measure vital signs using Doppler radars [30], [31], FMCW radars [32], or in time-domain UWB [26] and mm-Wave radars [2]. However, the measuring of breathing rate requires advanced signal processing algorithms that will not be necessary, in the event that the objective is only to detect the occupation, as proposed in this work. Fig. 13 compares the point cloud corresponding to two scenarios: the driver breathing normally or holding his breath during the acquisition time. In both cases, the person remains still. It can be observed

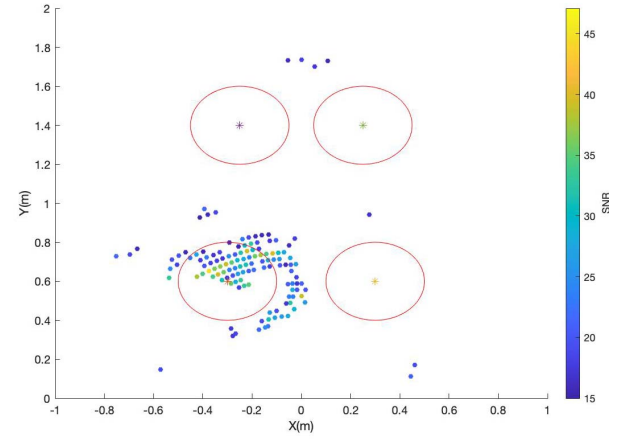


Fig. 14. Point cloud measured with the driver present and the car in motion.

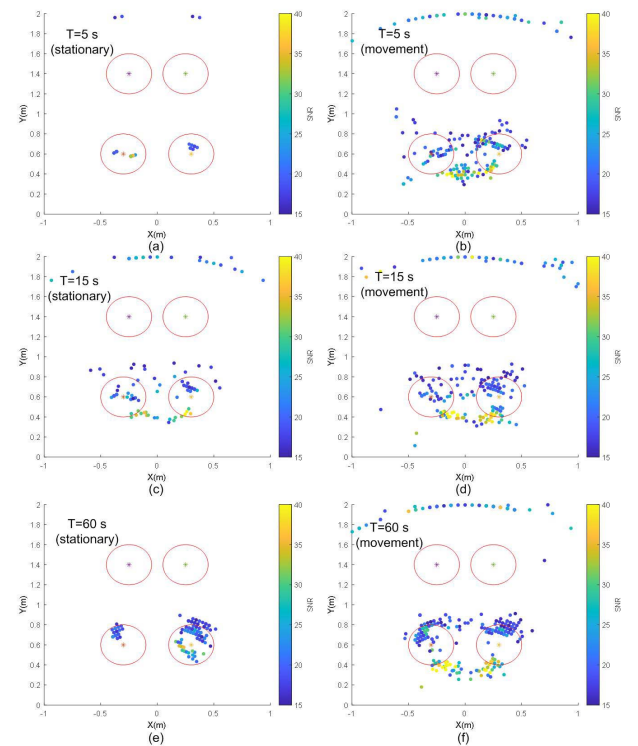


Fig. 15. Point cloud measured with the driver and passenger for different acquisitions periods (T_a): (a) with stationary persons (T_a) = 5s, (b) with the passenger in movement (T_a) = 5s, (c) with stationary persons (T_a) = 15s, (d) with the passenger in movement (T_a) = 15s, (e) with stationary persons (T_a) = 60s, and (f) with the passenger in movement (T_a) = 60s.

that the dispersion of the point cloud increases due to the movements of the chest, which are not completely filtered by the stationary clutter removal algorithm.

Another test was performed to show that vehicle movement does not significantly affect detection. Fig. 14 shows the results obtained; it can be observed that the cloud does not vary hardly with respect to the static case (i.e., the one shown in Fig. 11).

Fig. 15 shows the effects of the acquisition period T_a and the movement of the passengers on the point cloud. As the acquisition period increases, the total number of points N and the points in each region N_k increase. It also increases dispersion due to the inevitable movement of the occupants,

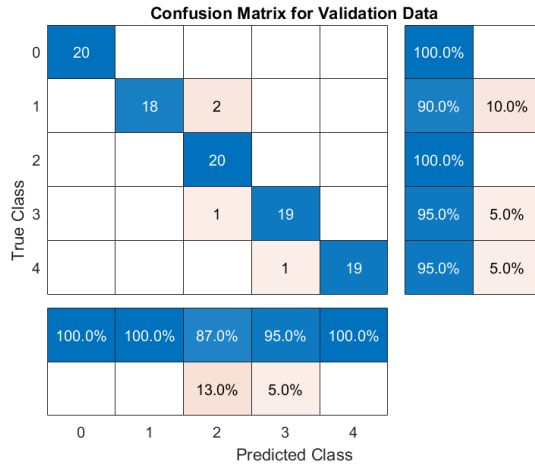


Fig. 16. Confusion matrix of the number of occupants for the case of a four-seater car.

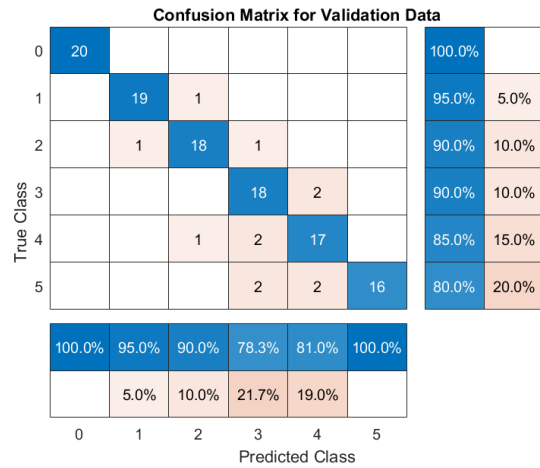


Fig. 17. Confusion matrix of the number of occupants for the case of a five-seater car.

despite remaining apparently still. Thus, in the figures on the left, the point cloud remains in the region of the seats. However, in the figures on the right, the passenger makes hand movements toward the driver to show these effects. It is observed that certain points outside the region of the seats correspond to such movements and to the presence of distant points produced by reflections in the vehicle cabin. The effect of the movement becomes greater dispersion σ_k . However, the movement does not increase the points associated with unoccupied seats (rear seats in this example). Longer acquisition times usually reduce the false alarms associated with random passenger movements, but, on the other hand, the occupancy rate is updated more slowly. Values of 10–15 s represent a compromise situation.

B. Classifier Performances

To validate the selection of the thresholds obtained from training samples, measurements were carried out with several volunteers. Different people sat in a vehicle in any of the seats at random, and the positions of the occupied seats were recorded manually and compared with the predictions given by the classification algorithm.

The confusion matrices of the number of occupants obtained for the case of four- and five-seat vehicles are shown in

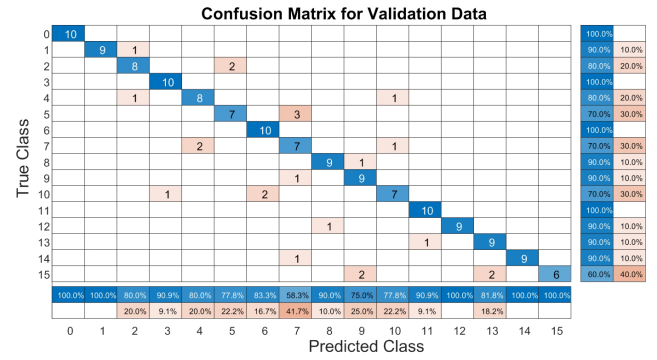


Fig. 18. Confusion matrix of seat occupancy status for a four-seater car.

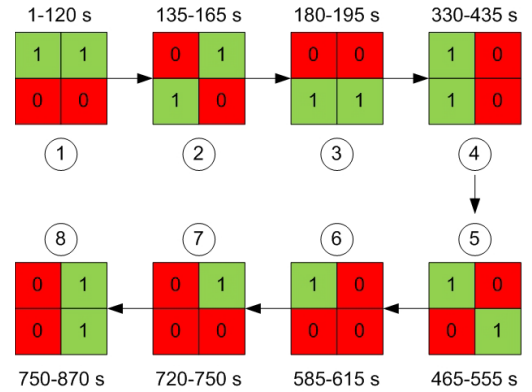


Fig. 19. Seat occupancy states as a function of time in seconds.

Figs. 16 and 17, respectively. In the confusion matrix plot, the rows are the predicted class (output class), and the columns are the true class (target class). In these figures, the class is the number of occupants in the car. A row summary shows the percentages of correctly and incorrectly classified observations for each true class. Similarly, a column summary shows the percentages of correctly and incorrectly classified observations for each predicted class. The accuracy found is 96% and 90% for four- and five-seats, respectively. The lower value obtained in the case of five seats is explained by the greater difficulty to distinguish the rear seats because the distance between seats is less than in the four-seater car.

In order to investigate the error in the prediction of the correct seat, Fig. 18 shows the confusion matrix of the state of each seat in the case of a four-seater car. The classes have been coded in binary (0001 seat number 1 occupied and 1111 for all seats occupied). It can be seen that most of the errors occur in the rear seats where the seating position is not as well defined as in the front ones. Even so, the error obtained is acceptable (accuracy of 86.125%), especially if the occupancy rate can be averaged over time, which will not vary as long as an occupant does not enter or exit the vehicle.

C. Dynamic Performance of the Occupancy Detector

The proposed algorithm has a low computational load; therefore, it can be implemented in real time, allowing the occupation to be dynamically updated. Figs. 19 and 20 show an example of occupancy detection (0 free and 1 occupied) of each seat of a four-seat car, as a function of time and the entry

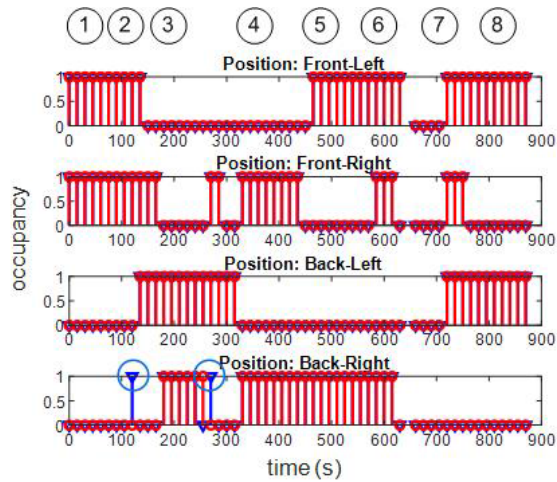


Fig. 20. Seat occupancy detected as a function of time. Marked with triangles circles, not smoothed, and with circles, smoothed. Circles in blue indicate false alarms during state transitions. The state number is indicated at the top of the figure corresponding to the cases described in Fig. 19.

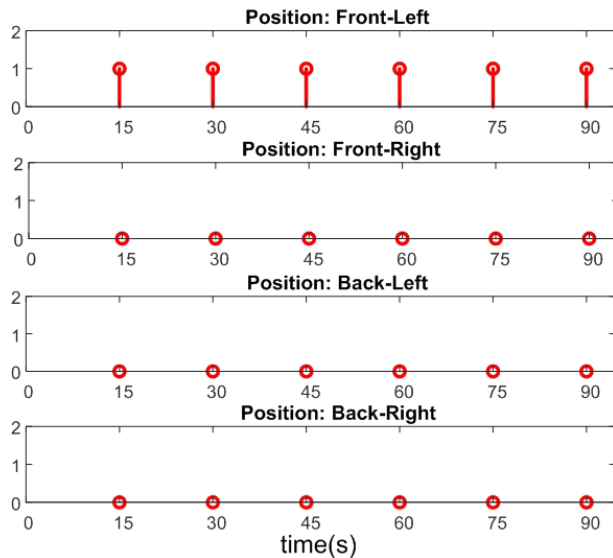


Fig. 21. Seat occupancy detected as a function of time for the case of a seat occupied by the driver performing hand movement over the seat on the side.

or exit of passengers from the vehicle. Fig. 19 describes the seats that are occupied in each time interval (state) identified with the number surrounded by a circle. Fig. 20 shows the occupancy detected as a function of the time for each of the time intervals described in Fig. 19. Some isolated errors can be observed, marked with blue circles, corresponding to the entry or exit of passengers from the vehicle (transitions represented in Fig. 19). However, the algorithm detects seat occupancy when the position remains stable. These failed points associated with transitions could be removed using the moving average filter provided in (16).

In order to investigate the effect of passenger movements, some experiments have been carried out. Fig. 21 shows the updated occupancy in periods of 15 s. In this case, the driver makes hand movements on the empty seat next to him. The cloud of points observed in Fig. 22 are the measurements

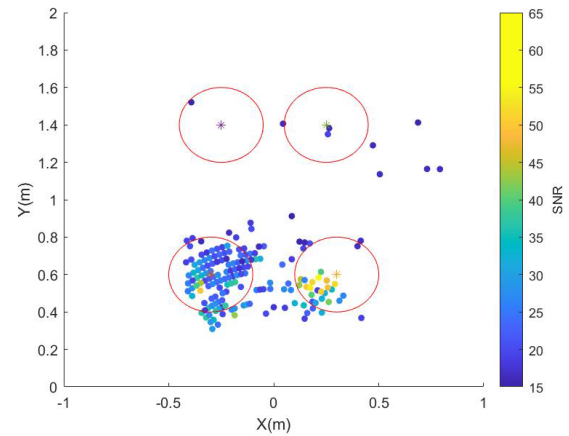


Fig. 22. Measured point cloud for the case of a seat occupied by the driver performing hand movement over the seat on the side.

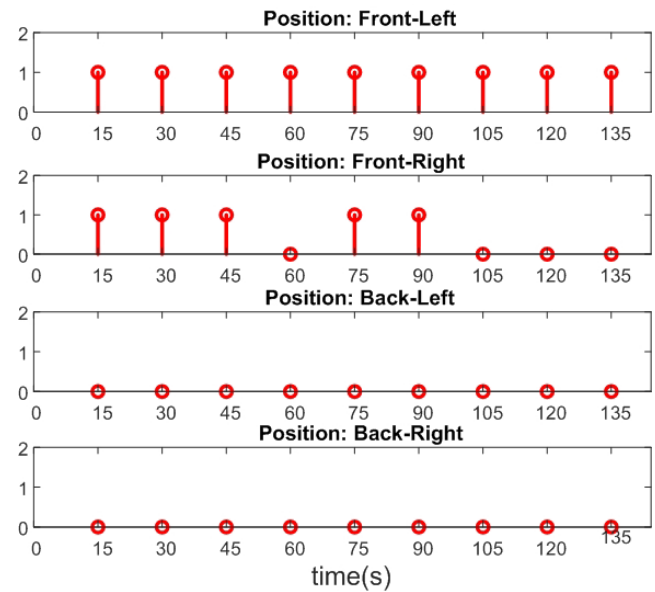


Fig. 23. Seat occupancy detected as a function of time for the case of a seat occupied by the driver performing large body movement over the seat on the side.

performing in the region of the seat where the hand passed. However, the algorithm is not able to detect the presence of seat occupancy in the empty seat because the points' density is relatively small. On the contrary, Fig. 23 shows the case in which the driver makes sudden movements invading the area of the adjacent seat. It can be seen how the algorithm detects the occupation of the side seat, while its invasion occurs. Fig. 24 shows the measured point cloud caused by the movement of the body on the lateral seat. However, the algorithm again detects the free seat in the absence of movement. Therefore, the proposed technique produces a false alarm in the presence of relevant passenger movements but returns to its steady state once these movements cease. These random false alarms can be detected if they occur when the car doors are closed, where the total occupancy should remain constant over time and can be considered outliers. These situations can be determined by combining pressure sensors (e.g., airbag sensors) in the seats that would indicate that there are no objects on the seat.

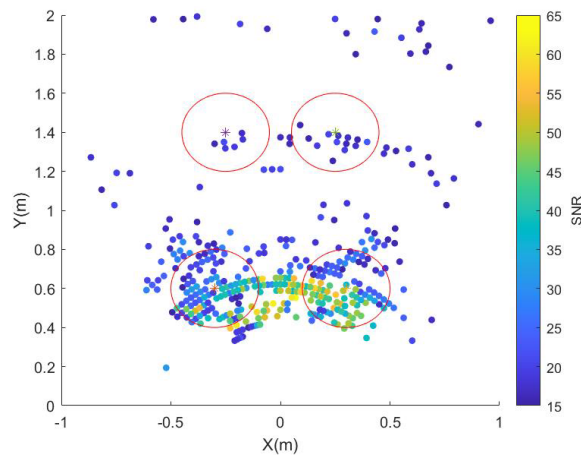


Fig. 24. Measured point cloud for the case of a seat occupied by the driver performing large body movement over the seat on the side.

VI. CONCLUSION AND FUTURE WORK

A vehicle occupancy detection system based on an MIMO FMCW mm-Wave radar at 77 GHz has been investigated. The radar is suspended on the ceiling of the vehicle although it can also be in other places where it has visibility of the passengers. An algorithm to determine seat occupancy from the measured point cloud read by the radar has been proposed. A static clutter removal algorithm is applied to consider only points associated with small body movements. Therefore, the system is able to detect and distinguish a passenger from other objects that may occupy the seat. A classification method is used, which is based on a function that takes into account the dispersion of the point cloud around the area in which the seat is located, as well as the point density of the cloud. The proposed algorithm needs few parameters that can be adjusted from training measurements. It has been observed that these values generally hold for different types of vehicles. Therefore, complex learning tools that require training procedures are not needed. These parameters have been manually adjusted. However, the fact that the number of returned points and the different signatures in the standard deviation for the occupied seats can be exploited in future works with advanced classifiers based on machine learning techniques. Although its implementation in embedded systems is not as easy as the proposed technique, the performance could be improved, and the results could be generalized to other types of vehicles or situations. Experimental results have been presented for four- and five-seater cars achieving high accuracy both in determining the number of occupants and the position of the occupied seat. This work opens the door to use the proposed occupancy detector combined with future vehicular communications systems, to apply it, for instance, to HOV lanes.

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